

Phase 3: Project Design Phase

AI Brand Voice & Marketing Content Generator

1. System Architecture Overview

The application is structured around a classic **3-Tier Architecture** model to separate presentation, business logic, and data layers. This decouples user interaction from resource-intensive API operations and persistence layers:

- **Presentation Layer (Streamlit):** Houses modular, responsive controls, visual progress states, and styling blocks.
- **Business Logic Layer (Python / Gemini API):** Orchestrates style extraction, few-shot generation, and iterative refinements.
- **Data Layer (SQLite):** Enforces 6 relational tables to persist user sessions, historical uploads, and content outputs.

2. Database Schema (Entity-Relationship Model)

The database design consists of 6 primary entities. It features rigid primary keys (PK), foreign keys (FK) for relationships, and cascading constraints to ensure complete integrity:

Table Name	Primary Key	Foreign Keys	Attributes & Relation Purpose
USER_ACCOUNT	UserID	None	UserName, Email, PasswordHash. Handles user registration and session profiles.
BRAND_PROFILE	BrandID	UserID (FK)	BrandName, CampaignPurpose. Represents a client's core marketing context.
SAMPLE_TEXT	SampleID	BrandID (FK), UserID (FK)	RawTextContent, DateUploaded. Stores raw copy used to extract style patterns.
BRAND_VOICE_LEARNING	LearningID	BrandID (FK)	SystemPromptTemplate, ExtractedTone, ExtractedVocabulary, EmotionalRhythm. Stores learned brand DNA.
CONTENT_REQUEST	RequestID	BrandID (FK), UserID (FK)	ContentType, GenerationTimestamp. Caches request configurations.
GENERATED_CONTENT	OutputID	RequestID (FK)	GeneratedText, CopyStatus. Stores raw generated text from Google Gemini.
CONTENT_REFINEMENT	RefinementID	OutputID (FK), RequestID (FK)	FeedbackInstruction, RegeneratedText, EditTimestamp. Manages revision versions.

3. Interface Layout & Sidebar Blueprint

The user interface layout is designed to maximize usability and content density:

- **Control Sidebar (Left 30% Width):** Contains configuration parameters. Inputs for Brand Name, Campaign Purpose, Brand Voice Samples (text area for learning), Content Format selector (dropdown containing Social Media Post, Marketing Email, Tagline, Ad Headline, Blog Intro), and a prominent 'Generate Content' trigger button.
- **Main Canvas (Right 70% Width):** Renders a modern, gradient header with modular cards. The bottom display area features a custom-tinted text block displaying generated content. Adjacent is an interactive feedback section containing a suggestion box and an 'Apply Changes' button, and a physical clipboard-copy utility.